FORCES

What is Force?

- A force is a push or pull on an object.
- Forces cause an object to accelerate…
  - To speed up
  - To slow down
  - To change direction

Units of force

- Newton (SI system)
  - 1 N = 1 kg *m/s^2
- 1 N is the force required to accelerate a 1 kg mass at a rate of 1 m/s^2
- Pound (British system)
  - 1 lb = 1 slug ft /s^2

Mass and Inertia

- Physicists define mass as inertia, which is the ability of a body to resist acceleration by a net force.
- Mass and inertia are directly related.
- Just like mass, the inertia of an object doesn’t change whether it is at rest or in motion.

Sample Problem

A heavy block hangs from a string attached to a rod. An identical string hangs down from the bottom of the block. Which string breaks

a) when the lower string is pulled with a slowly increasing force?

Top string breaks due to its greater force.

b) when the lower string is pulled with a quick jerk?

Bottom string breaks because block has lots of inertia and resists acceleration. Pulling force doesn't reach top string.

Gravity

A very common accelerating force is gravity. Here is gravity in action. The acceleration is *g*.

Mass and Weight

Mass and Weight

- Many people think mass and weight are the same thing. They are not.
- Mass is inertia, or resistance to acceleration.
- Weight can be defined as the force due to gravitational attraction.
- $w = mg$

Sample Problem

A man weighs 150 pounds on earth at sea level. Calculate his

a) mass in kg.

b) weight in Newtons.

Weight

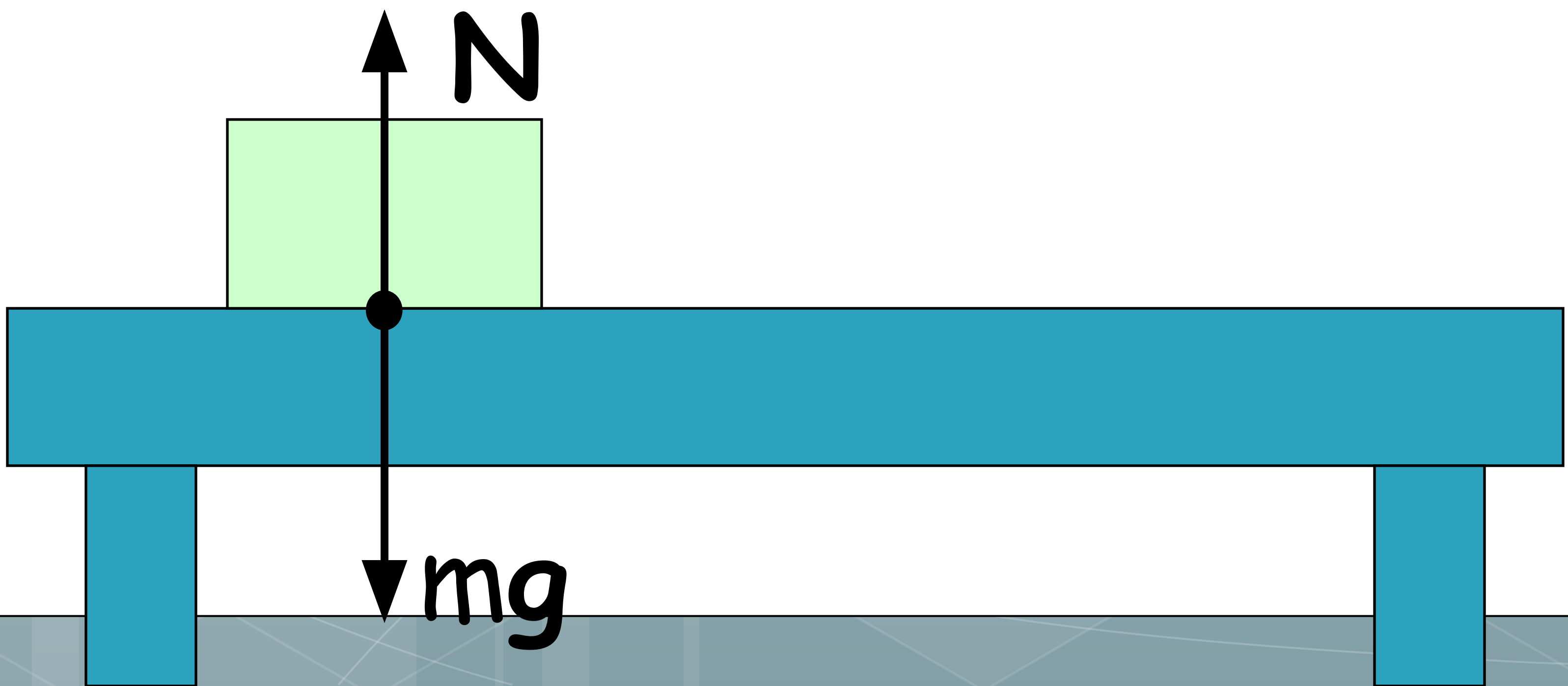
- What is the weight of a 30 kg object?
- w = 30 * 9.8 = 294 N

- What is the mass of a 490 N object?
- m = w/g = 490/9.8 = 50 kg

Normal Force

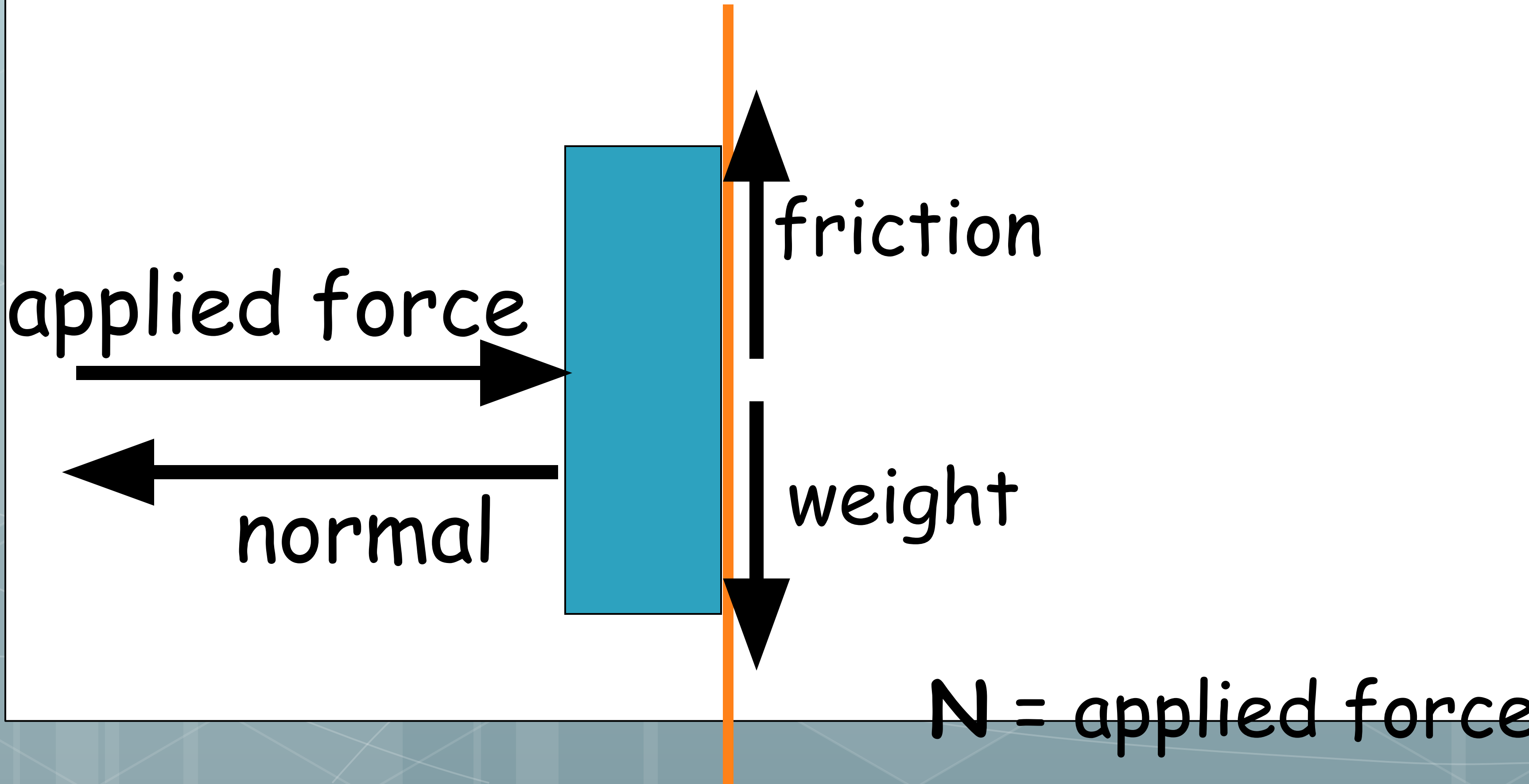
Normal force on flat surface

- The normal force is equal to the weight of an object for objects resting on ***horizontal*** surfaces.
- **N** = **w** = m**g**

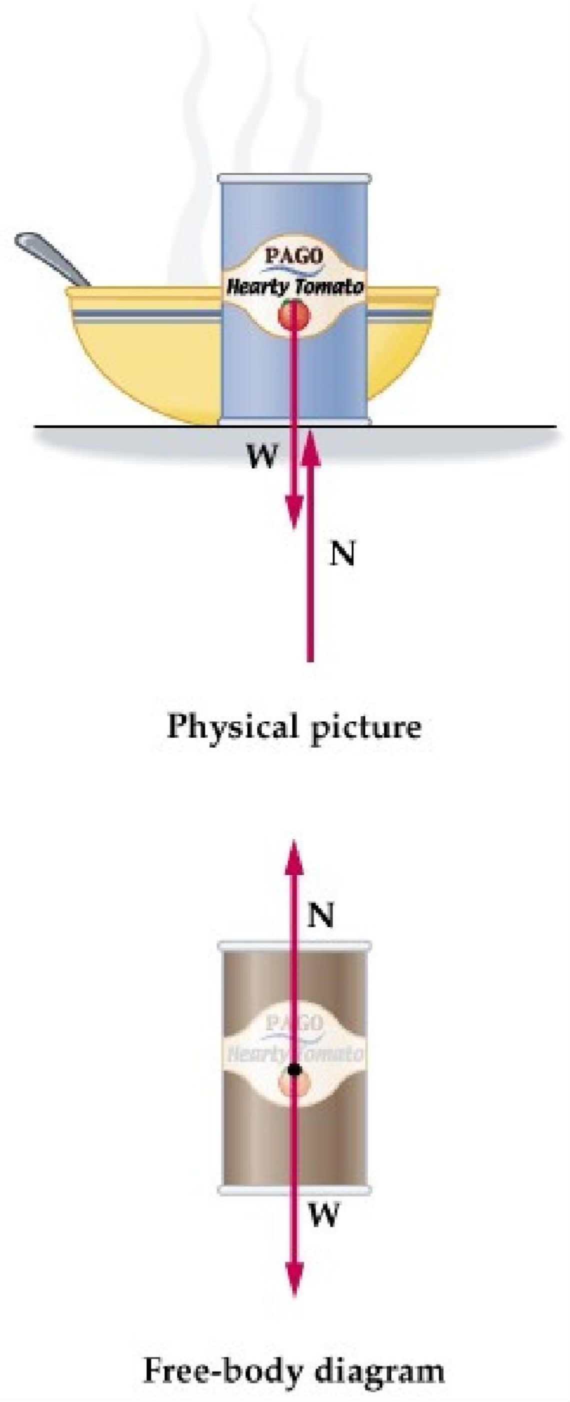


Normal force not associated with weight.

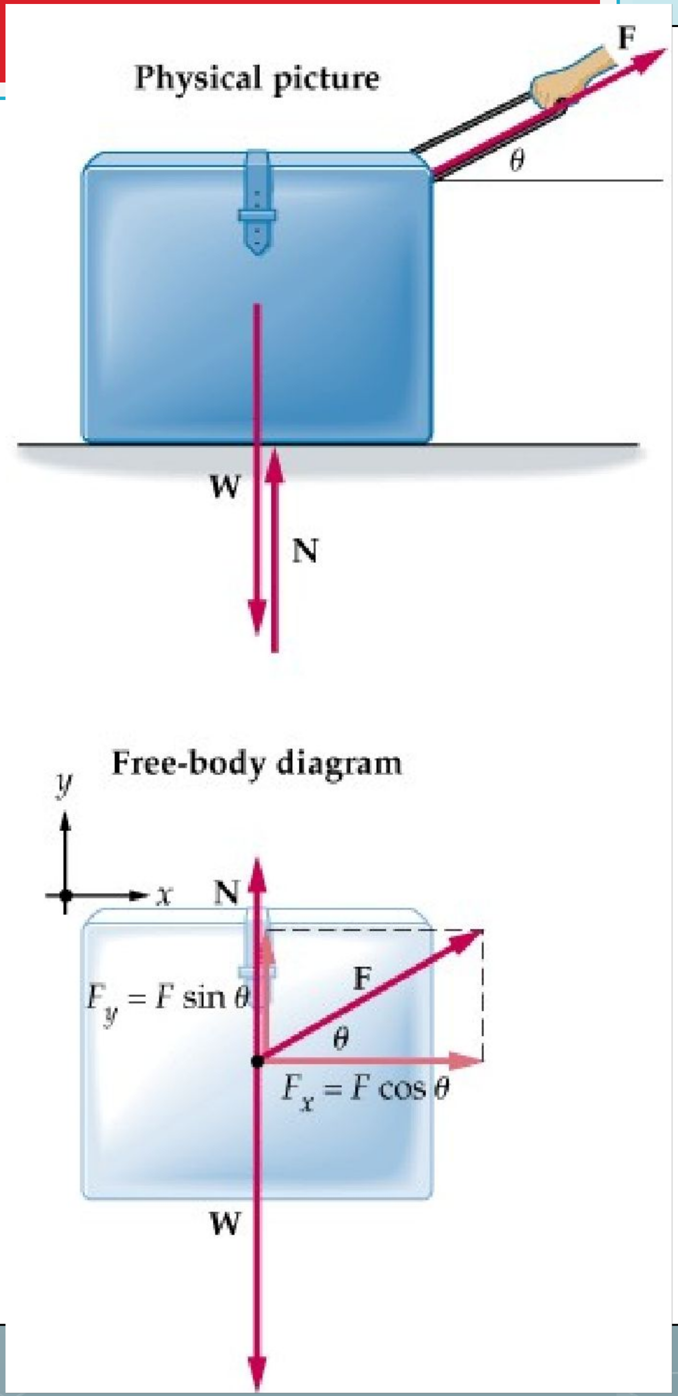
- A normal force can exist that is totally unrelated to the weight of an object.



The normal force most often equals the weight of an object…

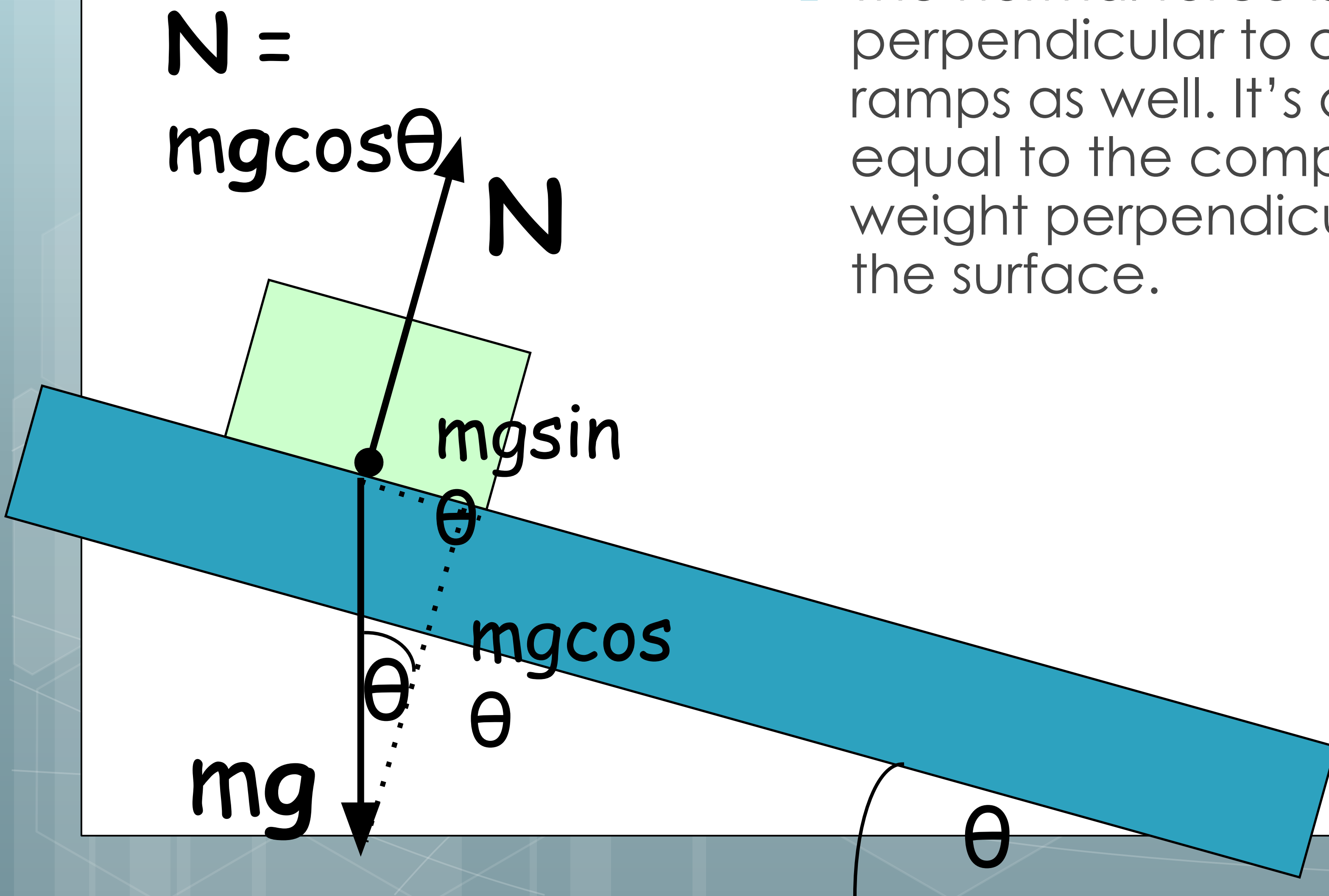


….but this is by no means always the case!



More on the Normal Force

Normal force on ramp



- The normal force is perpendicular to angled ramps as well. It’s always equal to the component of weight perpendicular to the surface.

Draw a free body diagram for the skier.

Sample problem

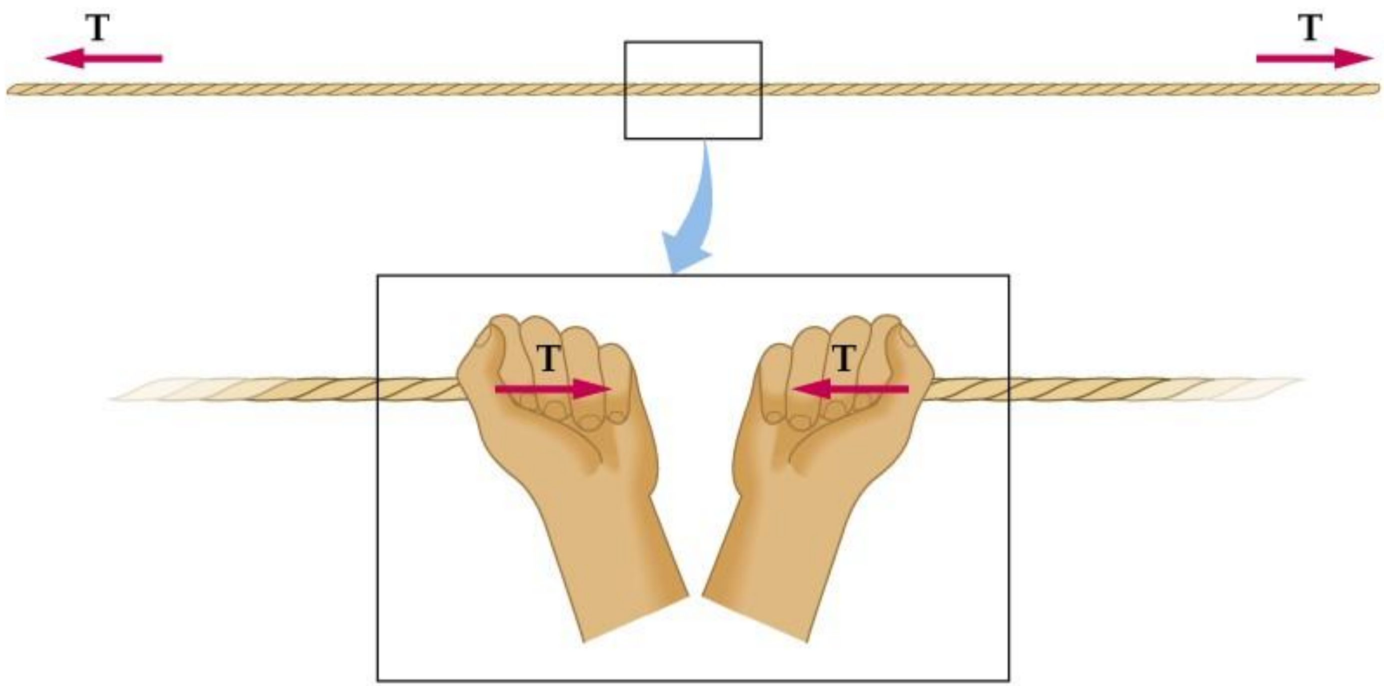
a) Find the normal force exerted on a 2.5-kg book resting on a surface inclined at 28° above the horizontal.

b) If the angle of the incline is reduced, do you expect the normal force to increase, decrease, or stay the same?

Tension

- Tension is a pulling force that arises when a rope, string, or other long thin material resists being pulled apart without stretching significantly.
- Tension always pulls away from a body attached to a rope or string and toward the center of the rope or string.

A physical picture of tension



Imagine tension to be the internal force preventing a rope or string from being pulled apart. Tension as such arises from the center of the rope or string. It creates an equal and opposite force on objects attached to opposite ends of the rope or string

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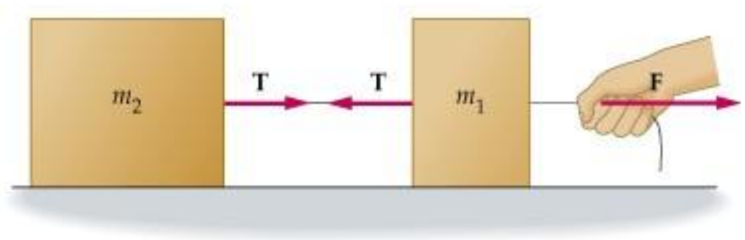
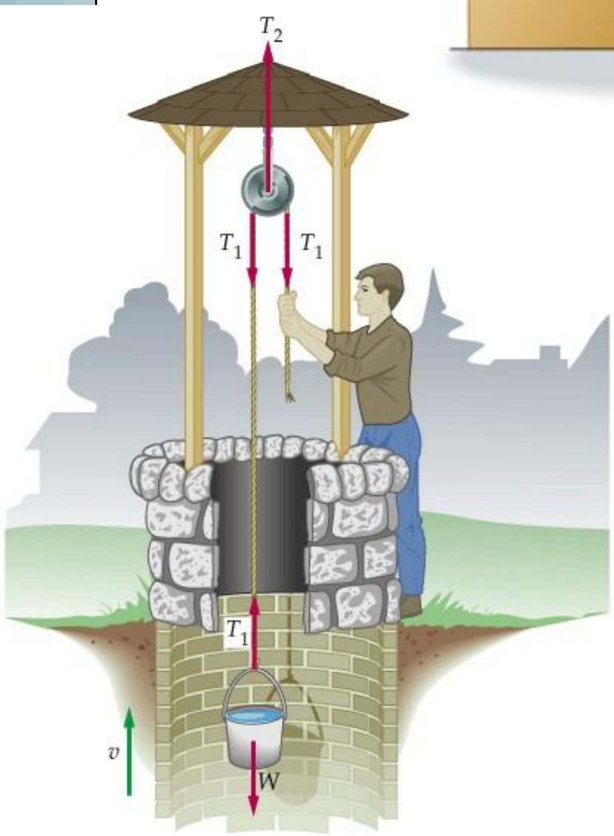
Tension examples

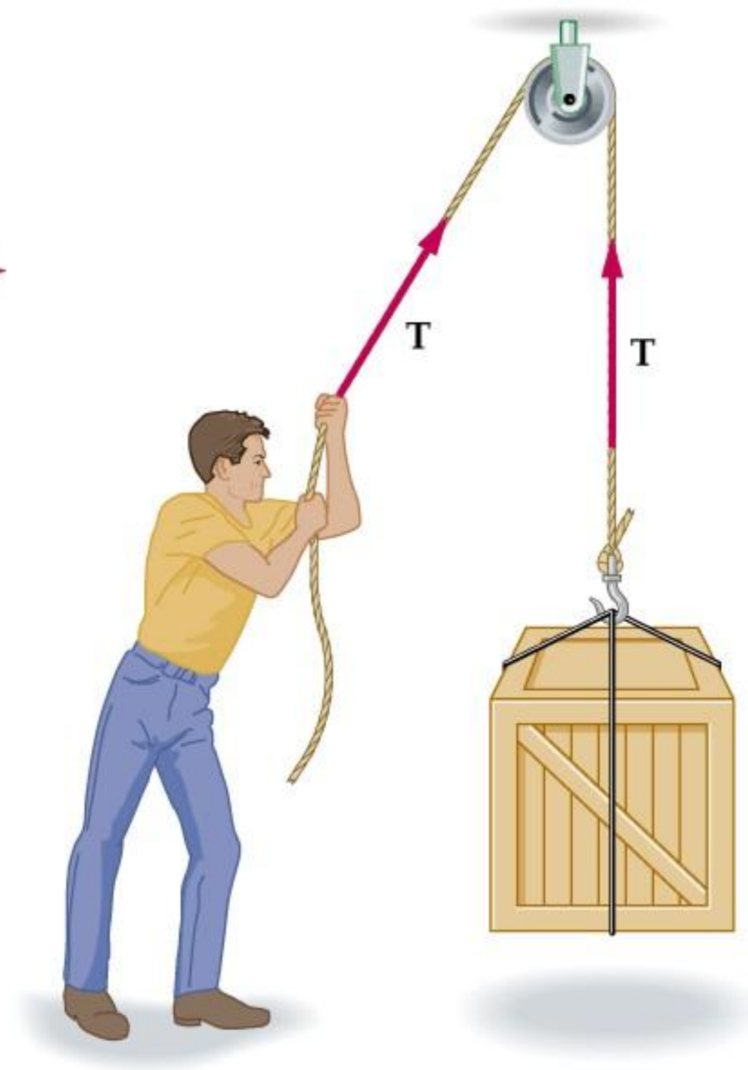


Physical picture



Physical picture

Note that the pulleys shown are magic! They affect the tension in any way, and serve only to bend the line of action of the force.



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Sample problem

A. A 1,500 kg crate hangs motionless from a crane cable. What is the tension in the cable? Ignore the mass of the cable.

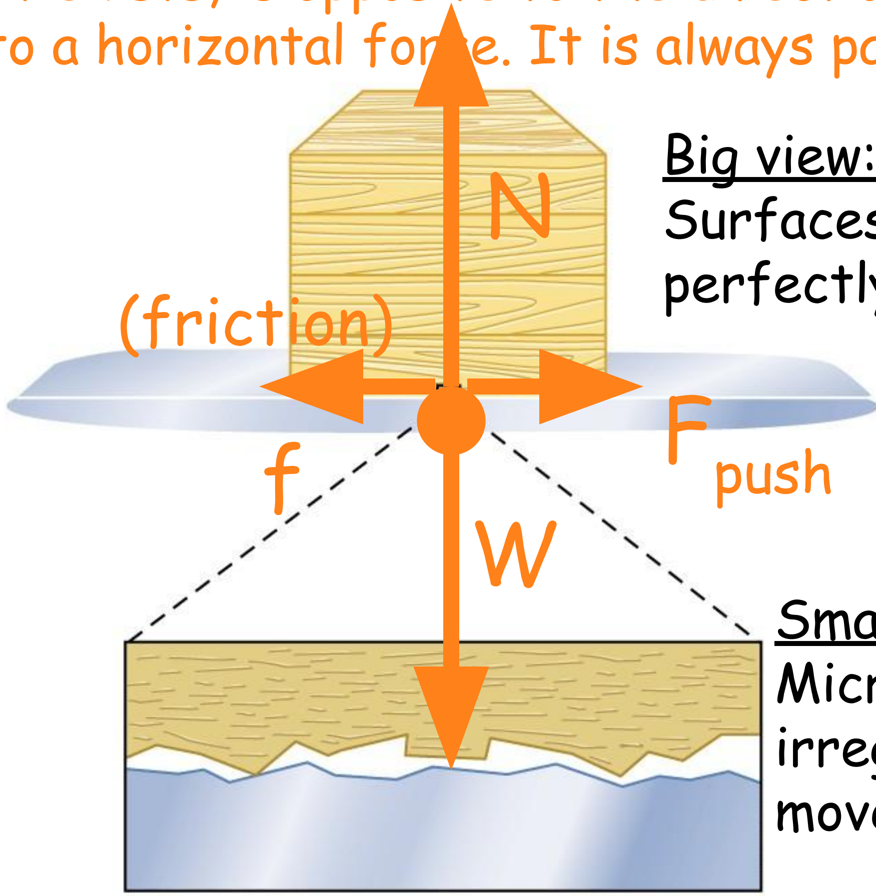
Introduction to Friction

Introduction to Friction

Friction

- Friction is the force that opposes a sliding motion.
- Friction is due to microscopic irregularities in even the smoothest of surfaces.
- Friction is highly useful. It enables us to walk and drive a car, among other things.
- Friction is also dissipative. That means it causes mechanical energy to be converted to heat. We'll learn more about that later.

Friction may or may not exist between two surfaces. The direction of friction, if it exists, is opposite to the direction object will slide when subjected to a horizontal force. It is always parallel to the surface.



<u>Big view:</u>
Surfaces look perfectly smooth.

<u>Small view:</u>
Microscopic irregularities resist movement.

Friction depends on the normal force.

- The friction that exists between two surfaces is directly proportional to the normal force.
- Increasing the normal force increases friction; decreasing the normal force decreases friction.

Static Friction

- This type of friction occurs between two surfaces that are not slipping relative to each other.
- $f_s \leq \mu_s N$
  - f_s : static frictional force (N)
  - μ_s: coefficient of static friction
  - N: normal force (N)

$f_s \leq \mu_s N$ *is an inequality!*

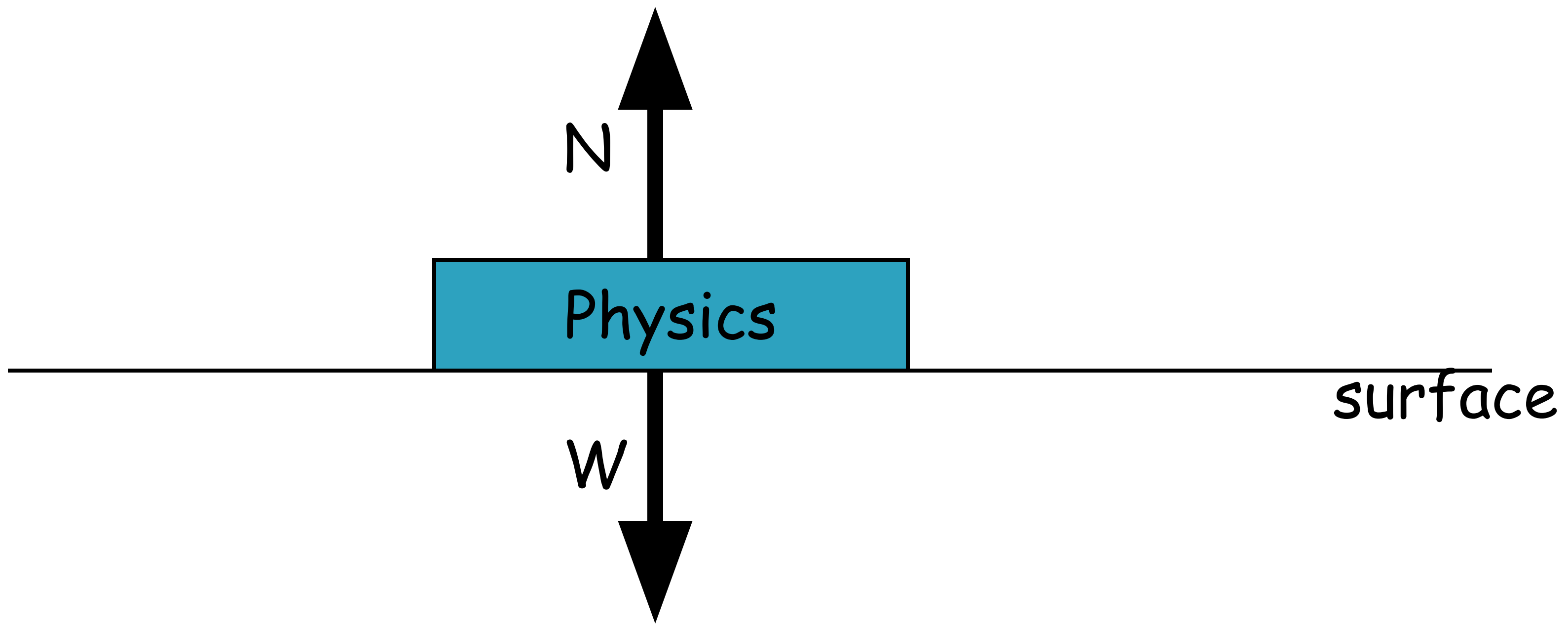
- The fact that the static friction equation is an inequality has important implications.
- Static friction between two surfaces is zero unless there is a force trying to make the surfaces slide on one another.
- Static friction can increase as the force trying to push an object increases until it reaches its maximum allowed value as defined by μ_s.
- Once the maximum value of static friction has been exceeded by an applied force, the surfaces begin to slide and the friction is no longer static friction.

Static friction and applied horizontal force

$f_s = 0$

There is no static friction since there is no applied horizontal force trying to slide the book on the surface.

Force Diagram

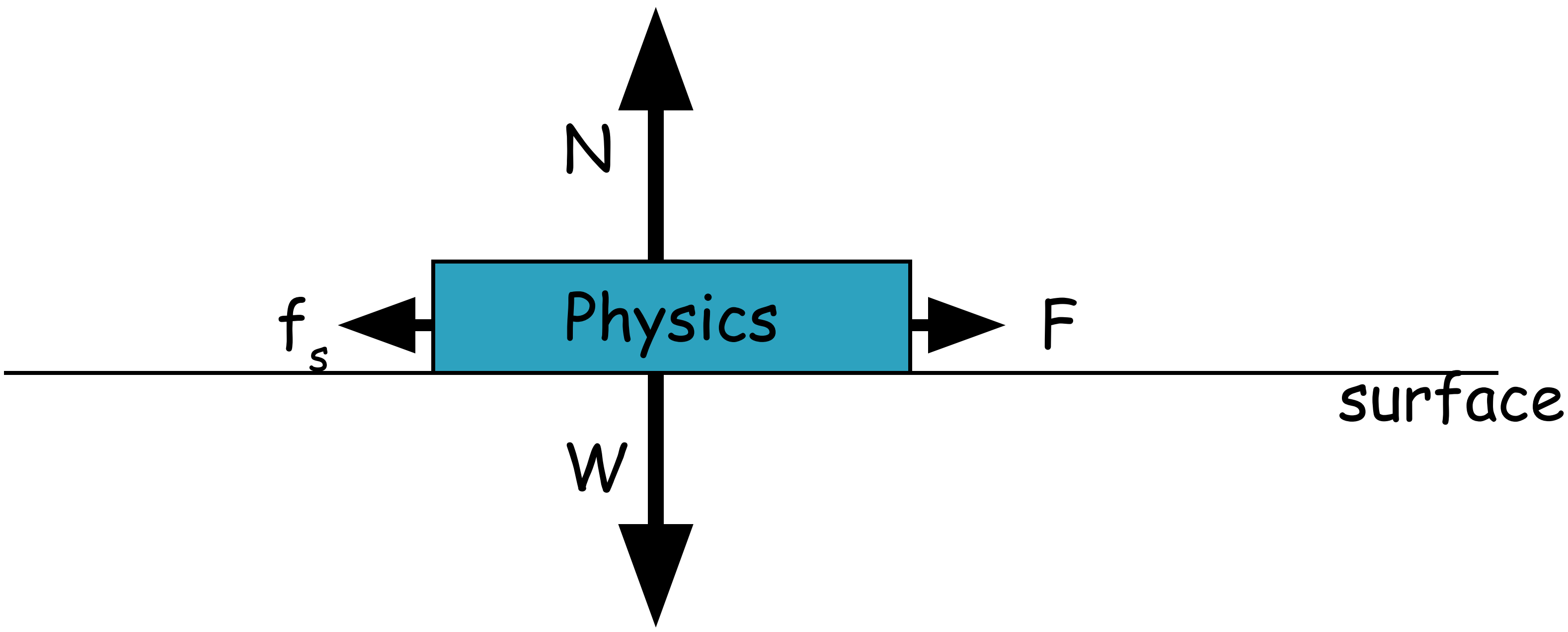


Static friction and applied horizontal force

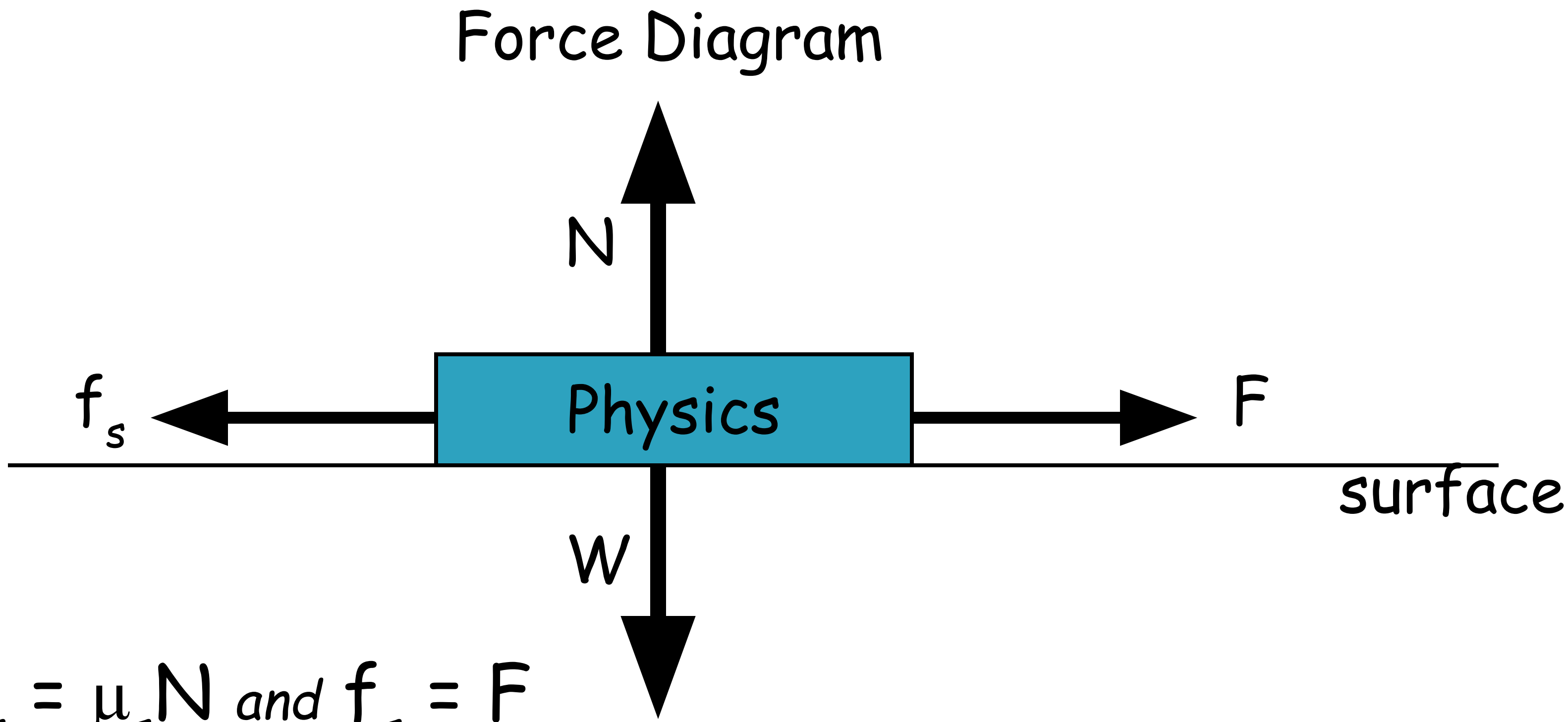
$0 < f_s < \mu_s N$ *and* $f_s = F$

Static friction is equal to the applied horizontal force, and there is no movement of the book since $\Sigma F = 0$.

Force Diagram



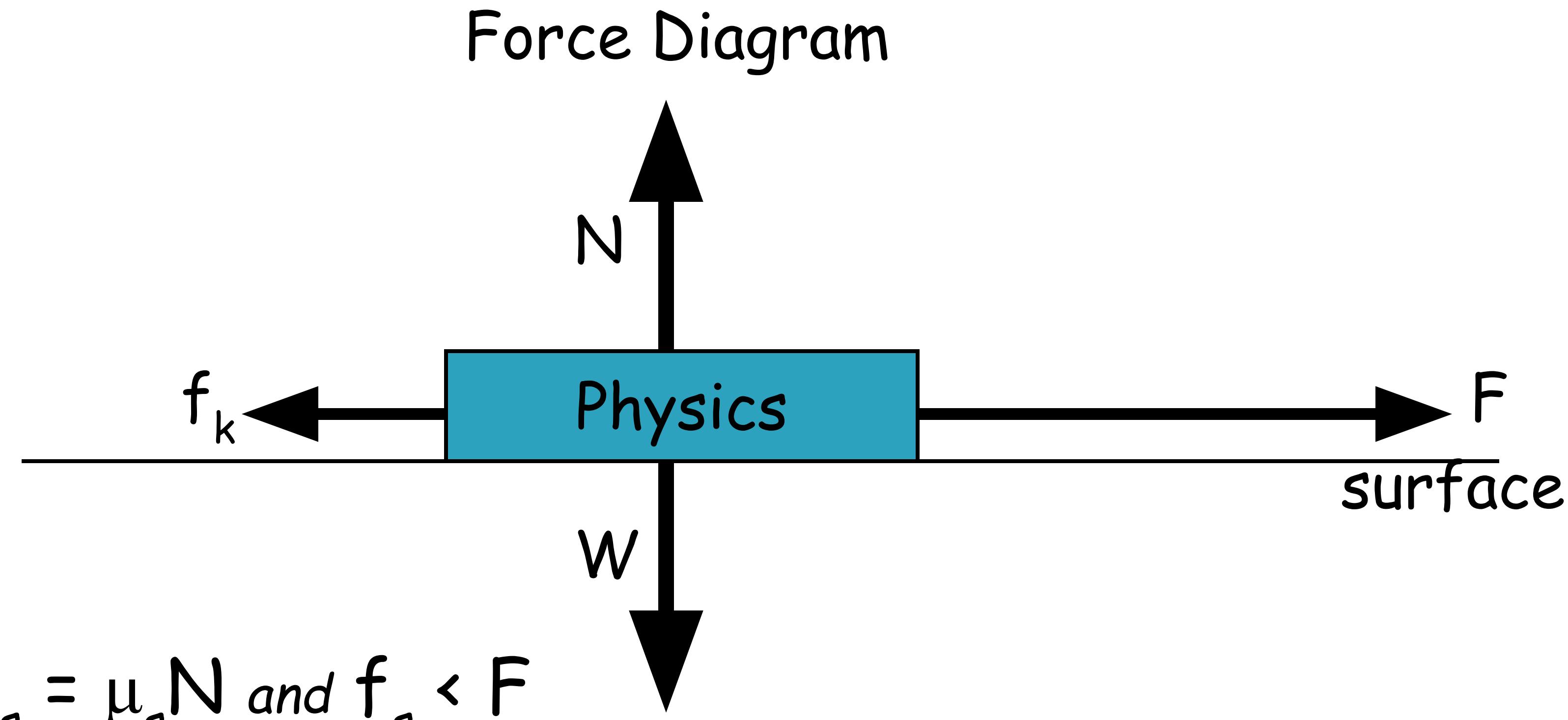
Static friction and applied horizontal force



$f_s = \mu_s N$ *and* $f_s = F$

Static friction is at its maximum value! It is still equal to F, but if F increases any more, the book will slide.

Static friction and applied horizontal force



$f_s = \mu_s N$ *and* $f_s < F$

Static friction cannot increase any more! The book accelerates to the right. Friction becomes kinetic friction, which is usually a smaller force.

Kinetic Friction

- This type of friction occurs between surfaces that are slipping past each other.
- $f_k = \mu_k N$
  - f_k : kinetic frictional force (N)
  - μ_k: coefficient of kinetic friction
  - N: normal force (N)
- Kinetic friction (sliding friction) is generally less than static friction (motionless friction) for most surfaces.

Sample Problem

A 10-kg box rests on a ramp that is laying flat. The coefficient of static friction is 0.50, and the coefficient of kinetic friction is 0.30.

a) What is the maximum horizontal force that can be applied to the box before it begins to slide? **50N**

b) What force is necessary to keep the box sliding at constant velocity?

30N

Sample Problem

A 10-kg wooden box rests on a board that is lying flat. The coefficient of static friction is 0.50, and the coefficient of kinetic friction is 0.30. What is the friction force between the box and ramp if

a) no force horizontal force is applied to the box? **0N**

b) a 20 N horizontal force is applied to the box? **20N**

c) a 60 N horizontal force is applied to the box? **30N**